

Functional Equations Workshop

Daniel, Lucy, Jeff, Brendan and Vir

UNSW Competitive Programming and Mathematics Society

October 7, 2026



Attendance

Please scan the QR code; please let us know about any dietary requirements (for the pizza!).



Functional Equation Basics

Acknowledgements

Questions here are from the following:

- BMO2 past papers from <https://bmos.ukmt.org.uk/>
- IMO shortlisted problems from <https://www.imo-official.org/>
- AoPS compilations of competition questions (e.g USAMO, Balkan Shortlist)
- Problem Solving Tactics by Angelo Di Pasquale, Norman Do, and Daniel Mathews
- Functional Equations in Mathematical Competitions: Problems and Solutions (2015) by Mohammad Mahid Taheri.
- OTIS Excerpts by Evan Chen, found on <https://web.evanchen.cc/textbooks/OTIS-Excerpts.pdf>

Acknowledgements

The workshop slides and lecture were written with the help of:

- OTIS Excerpts by Evan Chen, found on <https://web.evanchen.cc/textbooks/OTIS-Excerpts.pdf>
- Problem Solving Tactics by Angelo Di Pasquale, Norman Do, and Daniel Mathews
- functional equations (easy), found on <https://how-did-i-get-here.com/61/>
- Handouts from friends and AMT ran Math Camps.

Functions

- Functions take in a numerical input from a domain and gives out a numerical output from a codomain.
 - $f(x) = 2x$ is a function since any number passed through f will be doubled.

Functions

- Functions take in a numerical input from a domain and gives out a numerical output from a codomain.
 - $f(x) = 2x$ is a function since any number passed through f will be doubled.
 - $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x^2) = x$ isn't a function, as $f(1)$ could be either 1 or -1 .

Functions

- Functions take in a numerical input from a domain and gives out a numerical output from a codomain.
 - $f(x) = 2x$ is a function since any number passed through f will be doubled.
 - $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x^2) = x$ isn't a function, as $f(1)$ could be either 1 or -1 .
 - $g : \mathbb{R} \rightarrow \mathbb{R}^+$ where $g(x) = -x$ isn't a function, as $f(1) = -1$ is not a positive real,

Functions

- Functions take in a numerical input from a domain and gives out a numerical output from a codomain.
 - $f(x) = 2x$ is a function since any number passed through f will be doubled.
 - $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x^2) = x$ isn't a function, as $f(1)$ could be either 1 or -1 .
 - $g : \mathbb{R} \rightarrow \mathbb{R}^+$ where $g(x) = -x$ isn't a function, as $f(1) = -1$ is not a positive real,
- Functions are only defined if the input is in the domain. For example, if we have a function $f : \mathbb{N} \rightarrow \mathbb{N}$ where $f(x) = x$ for all positive integers x ,
 - $f(2) = 2$ as the value is the input

Functions

- Functions take in a numerical input from a domain and gives out a numerical output from a codomain.
 - $f(x) = 2x$ is a function since any number passed through f will be doubled.
 - $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x^2) = x$ isn't a function, as $f(1)$ could be either 1 or -1 .
 - $g : \mathbb{R} \rightarrow \mathbb{R}^+$ where $g(x) = -x$ isn't a function, as $f(1) = -1$ is not a positive real,
- Functions are only defined if the input is in the domain. For example, if we have a function $f : \mathbb{N} \rightarrow \mathbb{N}$ where $f(x) = x$ for all positive integers x ,
 - $f(2) = 2$ as the value is the input
 - $f(\pi)$ isn't defined as π isn't a positive integer.

Functions

- Functions take in a numerical input from a domain and gives out a numerical output from a codomain.
 - $f(x) = 2x$ is a function since any number passed through f will be doubled.
 - $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x^2) = x$ isn't a function, as $f(1)$ could be either 1 or -1 .
 - $g : \mathbb{R} \rightarrow \mathbb{R}^+$ where $g(x) = -x$ isn't a function, as $f(1) = -1$ is not a positive real,
- Functions are only defined if the input is in the domain. For example, if we have a function $f : \mathbb{N} \rightarrow \mathbb{N}$ where $f(x) = x$ for all positive integers x ,
 - $f(2) = 2$ as the value is the input
 - $f(\pi)$ isn't defined as π isn't a positive integer.

Functional Equations

Functional equations are equations which give us a property (or properties) of a function, along with the domain and codomain of the function. Examples include:

Functional Equations

Functional equations are equations which give us a property (or properties) of a function, along with the domain and codomain of the function. Examples include:

- Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all $x, y \in \mathbb{R}$:

$$f(x + y)^2 = f(x)^2 + f(y)^2$$

Functional Equations

Functional equations are equations which give us a property (or properties) of a function, along with the domain and codomain of the function. Examples include:

- Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all $x, y \in \mathbb{R}$:

$$f(x + y)^2 = f(x)^2 + f(y)^2$$

- Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all $x, y \in \mathbb{R}$:

$$f(xf(y)) + f(y) = f(x + y) + f(xy).$$

Functional Equations

Functional equations are equations which give us a property (or properties) of a function, along with the domain and codomain of the function. Examples include:

- Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all $x, y \in \mathbb{R}$:

$$f(x+y)^2 = f(x)^2 + f(y)^2$$

- Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all $x, y \in \mathbb{R}$:

$$f(xf(y)) + f(y) = f(x+y) + f(xy).$$

In general, the main parts of a functional equation question are:

- The name and defined domain and codomain of the function.
- The actual "functional equation".

and the objective is to find all functions satisfying the equation.

Injectivity and Surjectivity

Injectivity and Surjectivity

Injective Function

A function $f : A \rightarrow B$ is injective iff every unique input $a \in A$ produces a unique output $f(a)$. That is

$$f(x) = f(y) \implies x = y$$

for every $x, y \in A$.

Surjective Function

A function $f : A \rightarrow B$ is surjective iff for every $b \in B$, there exists some value $a \in A$ such that $f(a) = b$.

$$\forall b \in B, \exists a \in A : f(a) = b.$$

A function f is bijective if and only if it is both injective and surjective.

Example Problem

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real x, y ,

$$f(f(x) + y) = x + f(f(y)).$$

Example Problem

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real x, y ,

$$f(f(x) + y) = x + f(f(y)).$$

- Can we prove that f is injective?

Example Problem

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real x, y ,

$$f(f(x) + y) = x + f(f(y)).$$

- Can we prove that f is injective?
- Can we prove that f is surjective?

Let us first prove injectivity.

Assume there are some values $a, b \in \mathbb{R}$ such that $f(a) = f(b)$.

Example Problem

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real x, y ,

$$f(f(x) + y) = x + f(f(y)).$$

Assume there are some values $a, b \in \mathbb{R}$ such that $f(a) = f(b) = c$. Thus, by substituting a and b into x , we get:

$$f(f(a) + y) = a + f(f(y)),$$

$$f(f(b) + y) = b + f(f(y))$$

As $f(a) = f(b)$, the right-hand sides are equal, and hence the left-hand sides are also equal. Subtracting $f(f(y))$ from both sides gives $a = b$. Therefore, f is injective.

Example Problem

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real x, y ,

$$f(f(x) + y) = x + f(f(y)).$$

- Can we prove that f is surjective?

Let us now prove surjectivity.

Example Problem

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real x, y ,

$$f(f(x) + y) = x + f(f(y)).$$

- Can we prove that f is surjective?

Let us now prove surjectivity.

Consider some arbitrary value $k \in \mathbb{R}$. We now wish to show that there exists some $a \in \mathbb{R}$ such that $f(a) = k$.

Example Problem

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real x, y ,

$$f(f(x) + y) = x + f(f(y)).$$

Now, consider the substitution $(x', y') = (k - f(f(y)), y)$. We get:

$$\begin{aligned} f(f(x') + y) &= k - f(f(y)) + f(f(y)) \\ &= k. \end{aligned}$$

Thus, if we take

$$a = f(x') + y = f(k - f(f(y))) + y,$$

then $f(a) = k$, as desired.

Substitution

Substitution

If injectivity and surjectivity were bread and butter, this is the knife you use to spread it. Since functional equations hold true for all values within their domain, you are free to substitute any number/expression into the equation.

Example Problem

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real x, y ,

$$f(f(x) + y) = x + f(f(y)).$$

Consider the substitution $x = 0$. We find that:

$$f(f(0) + y) = f(f(y)).$$

Since the outputs of two values are equal and ***we know that the function is injective***, we find that:

$$f(0) + y = f(y)$$

for all reals $y \in \mathbb{R}$. Thus, we get that the function satisfies $f(y) = y + c$ for the constant $c = f(0)$.

Example Problem

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real x, y ,

$$f(f(x) + y) = x + f(f(y)).$$

- If $f(y) = y + c$, is it actually a solution?

By substituting $f(y) = y + c$ back into the functional equation, we find

$$\begin{aligned} f(x + y + c) &= x + f(y + c), \\ \implies x + y + 2c &= x + y + 2c \end{aligned}$$

which is true. Thus, all solutions are of the form $f(x) = x + c$ for a constant c .

How to Substitute

In general, consider substitutions that will make the FE look nicer, regardless of what the actual numbers substituted are.

How to Substitute

In general, consider substitutions that will make the FE look nicer, regardless of what the actual numbers substituted are.

SUB SUB SUB!

Common Substitutions

Here is a small list of common substitutions for functions.

Common Substitutions

Here is a small list of common substitutions for functions.

- Substituting in either $x = 0$, or $x = \pm 1$.

Common Substitutions

Here is a small list of common substitutions for functions.

- Substituting in either $x = 0$, or $x = \pm 1$.
- Substituting $x = y$, $x = \frac{1}{y}$, or $x = -y$

Common Substitutions

Here is a small list of common substitutions for functions.

- Substituting in either $x = 0$, or $x = \pm 1$.
- Substituting $x = y$, $x = \frac{1}{y}$, or $x = -y$
- Substituting x or y such that one bracket vanishes. For example, substituting $x = -1$ if there is $f(x + 1)$.

Common Substitutions

Here is a small list of common substitutions for functions.

- Substituting in either $x = 0$, or $x = \pm 1$.
- Substituting $x = y$, $x = \frac{1}{y}$, or $x = -y$
- Substituting x or y such that one bracket vanishes. For example, substituting $x = -1$ if there is $f(x + 1)$.
- Substituting x or y such that two brackets become equal. For example, substituting $y = 2x$ if there is $f(3x)$ and $f(x + y)$.

Common Substitutions

Here is a small list of common substitutions for functions.

- Substituting in either $x = 0$, or $x = \pm 1$.
- Substituting $x = y$, $x = \frac{1}{y}$, or $x = -y$
- Substituting x or y such that one bracket vanishes. For example, substituting $x = -1$ if there is $f(x + 1)$.
- Substituting x or y such that two brackets become equal. For example, substituting $y = 2x$ if there is $f(3x)$ and $f(x + y)$.
- When the domain is an integer, try small values $(2, 3, 4, \dots)$.

Common Substitutions

Here is a small list of common substitutions for functions.

- Substituting in either $x = 0$, or $x = \pm 1$.
- Substituting $x = y$, $x = \frac{1}{y}$, or $x = -y$
- Substituting x or y such that one bracket vanishes. For example, substituting $x = -1$ if there is $f(x + 1)$.
- Substituting x or y such that two brackets become equal. For example, substituting $y = 2x$ if there is $f(3x)$ and $f(x + y)$.
- When the domain is an integer, try small values $(2, 3, 4, \dots)$.
- If f is linear, try $f(x) = ax + b$ for real constants a and b .

Common Substitutions

Here is a small list of common substitutions for functions.

- Substituting in either $x = 0$, or $x = \pm 1$.
- Substituting $x = y$, $x = \frac{1}{y}$, or $x = -y$
- Substituting x or y such that one bracket vanishes. For example, substituting $x = -1$ if there is $f(x + 1)$.
- Substituting x or y such that two brackets become equal. For example, substituting $y = 2x$ if there is $f(3x)$ and $f(x + y)$.
- When the domain is an integer, try small values $(2, 3, 4, \dots)$.
- If f is linear, try $f(x) = ax + b$ for real constants a and b .
- Sometimes it helps to consider comparing similar substitutions of x .

Common Substitutions

Here is a small list of common substitutions for functions.

- Substituting in either $x = 0$, or $x = \pm 1$.
- Substituting $x = y$, $x = \frac{1}{y}$, or $x = -y$
- Substituting x or y such that one bracket vanishes. For example, substituting $x = -1$ if there is $f(x + 1)$.
- Substituting x or y such that two brackets become equal. For example, substituting $y = 2x$ if there is $f(3x)$ and $f(x + y)$.
- When the domain is an integer, try small values $(2, 3, 4, \dots)$.
- If f is linear, try $f(x) = ax + b$ for real constants a and b .
- Sometimes it helps to consider comparing similar substitutions of x .
- Create a transformation of the function (e.g. $g(x) = (x - 1)f(x)$).

Example Problem

Substitution Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f\left(\frac{x+y}{2}\right) = f(f(x) + y) - \frac{1}{4}y.$$

Example Problem

Substitution Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f\left(\frac{x+y}{2}\right) = f(f(x) + y) - \frac{1}{4}y.$$

Motivation: Get rid of fraction and try forming relationship in terms of $f(f(x) + y)$ only.

Example Problem

Substitution Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f\left(\frac{x+y}{2}\right) = f(f(x) + y) - \frac{1}{4}y.$$

Motivation: Get rid of fraction and try forming relationship in terms of $f(f(x) + y)$ only.

Let us equate $\frac{x+y}{2}$ with $f(x) + y$, giving us $y = x - 2f(x)$.

Example Problem

Substitution Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f\left(\frac{x+y}{2}\right) = f(f(x) + y) - \frac{1}{4}y.$$

Motivation: Get rid of fraction and try forming relationship in terms of $f(f(x) + y)$ only.

Let us equate $\frac{x+y}{2}$ with $f(x) + y$, giving us $y = x - 2f(x)$.

Using this substitution yields

$$f(x - f(x)) = f(x - f(x)) - \frac{1}{4}(x - 2f(x))$$

$$x - 2f(x) = 0$$

$$f(x) = \frac{x}{2}.$$

Example Problem: Shoving in Numbers

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y) + x) = xy + f(x)$$

for all $x, y \in \mathbb{R}$.

Example Problem

First, let's substitute $x = 1$ into the equation and let $a = f(1)$. Then, we get that

$$f(f(y) + 1) = y + a$$

and from this we claim that f is injective. Indeed, if $f(x) = f(y)$, then

$$f(f(x) + 1) = f(f(y) + 1)$$

$$\implies x + a = y + a$$

$$\implies x = y$$

and so f is injective. Now, we substitute $x = 1, y = 0$ into the original equation, and we get that

$$f(f(0) + 1) = f(1),$$

which implies $f(0) = 0$ by injectivity of f .

Example Problem

If we substitute $x = 1, y = -a$ then by the injectivity of f we get that

$$\begin{aligned} f(f(-a) + 1) &= 0 = f(0) \\ \implies f(-a) &= -1, \end{aligned}$$

and so, substituting only $y = -a$ gives us that

$$\begin{aligned} f(f(-a)x + x) &= -ax + f(x) = f(0) = 0 \\ \implies f(x) &= ax. \end{aligned}$$

Substituting $f(x) = ax$ into the original equation yields

$$a^2xy + ax = xy + ax,$$

meaning that $a = \pm 1$. Hence $f(x) = \pm x$.

Additive Functions

Additive and Multiplicative

Definition

A function $f : A \rightarrow B$ is additive if and only if for every $x, y \in A$:

$$f(x + y) = f(x) + f(y).$$

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

- Maybe we should try some small cases: find $f(1), f(2), \dots$

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

- Maybe we should try some small cases: find $f(1), f(2), \dots$

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

- Maybe we should try some small cases: find $f(1), f(2), \dots$

Let $f(1) = 1$ (we will change this later). What will $f(2)$ be?

$$f(2) = f(1) + f(1) = 1 + 1 = 2.$$

Then what will $f(3)$ be?

$$f(3) = f(2) + f(1) = 2 + 1 = 3.$$

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

- Maybe we should try some small cases: find $f(1), f(2), \dots$

Let $f(1) = 1$ (we will change this later). What will $f(2)$ be?

$$f(2) = f(1) + f(1) = 1 + 1 = 2.$$

Then what will $f(3)$ be?

$$f(3) = f(2) + f(1) = 2 + 1 = 3.$$

We can conjecture that $f(n) = n$ for all positive integers n .

How would we prove this?

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

- “We can conjecture that $f(n) = n$ for all positive integers n .” How would we prove this?

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

- “We can conjecture that $f(n) = n$ for all positive integers n .” How would we prove this?

We can induct on n . We've assumed that $f(1) = 1$, so we only need to prove the inductive step.

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

- “We can conjecture that $f(n) = n$ for all positive integers n .” How would we prove this?

We can induct on n . We've assumed that $f(1) = 1$, so we only need to prove the inductive step.

Assume that $f(k) = k$ for some positive integer k . We can then see that

$$f(k + 1) = f(k) + f(1) = k + 1,$$

and thus $f(k + 1) = k + 1$, as desired.

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

- Now what happens if $f(1) = k$ for an arbitrary positive integer k instead?

Cauchy's Functional Equation

Cauchy's Functional Equation

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(x + y) = f(x) + f(y)$$

for all positive integers x, y .

- Now what happens if $f(1) = k$ for an arbitrary positive integer k instead?

We can use a similar argument to show that $f(n) = nk$ for all positive integers n . Thus, the general solution is

$$f(n) = nk$$

for some arbitrary fixed positive integer k .

Cauchy's Functional Equation

Remark: A function $f : \mathbb{Q} \rightarrow \mathbb{R}$ satisfying

$$f(x + y) = f(x) + f(y)$$

will always be of the form $f(x) = kx$ for some fixed real k .

The codomain doesn't *really* affect the proof, but the domain does!

Symmetry

Exploiting Symmetry

Example 1

Find all functions $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xg(y) + yg(x) = (x + y)g(x^2 + y^2)$$

for all $x, y \in \mathbb{R}$.

Example 2

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)) + f(f(y)) = f(x) + y$$

for all $x, y \in \mathbb{R}$.

Exploiting Symmetry

Example 1

Find all functions $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xg(y) + yg(x) = (x + y)g(x^2 + y^2)$$

for all $x, y \in \mathbb{R}$.

Example 2

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)) + f(f(y)) = f(x) + y$$

for all $x, y \in \mathbb{R}$.

What happens when we switch x and y ?

Case 1

Let's see what happens when we switch x and y in the first example.

Case 1

Let's see what happens when we switch x and y in the first example. After switching x and y , functional equation becomes

$$yg(x) + xg(y) = (y + x)g(y^2 + x^2),$$

which is the exact same as the previous equation. So swapping x and y does not help us to solve the problem.

Case 2

Now for the second example, let's see what happens when we switch x and y .

Case 2

Now for the second example, let's see what happens when we switch x and y . The equation becomes

$$f(f(y)) + f(f(x)) = f(y) + x,$$

so our equation has changed in at least one way. In particular, the LHS remains the same but the RHS has changed, so we can equate both equations to yield that

$$f(y) + x = f(x) + y.$$

Case 2

Now for the second example, let's see what happens when we switch x and y . The equation becomes

$$f(f(y)) + f(f(x)) = f(y) + x,$$

so our equation has changed in at least one way. In particular, the LHS remains the same but the RHS has changed, so we can equate both equations to yield that

$$f(y) + x = f(x) + y.$$

This is a lot easier to solve.

Exploiting Symmetry

So, what worked in the second example that didn't work in our first example?

Exploiting Symmetry

So, what worked in the second example that didn't work in our first example?

The first example is 'symmetric', so swapping x and y does not change the equation at all. The second example is 'almost symmetric', so the equation after swapping x and y 'agrees' with the original equation on almost everything except for a few parts. In particular, the parts where the equations do not agree give us a new equation which was much easier to solve.

Exploiting Symmetry

So, what worked in the second example that didn't work in our first example?

The first example is 'symmetric', so swapping x and y does not change the equation at all. The second example is 'almost symmetric', so the equation after swapping x and y 'agrees' with the original equation on almost everything except for a few parts. In particular, the parts where the equations do not agree give us a new equation which was much easier to solve.

Moral of the story: if there are multiple variables, try swapping these variables and see what happens!

Example Problem

Example

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x)f(y) = x + f(y) + 57$$

for all $x, y \in \mathbb{R}$.

Example Problem

First, swapping x and y gives us that

$$f(y)f(x) = y + f(x) + 57,$$

so the LHS remains the same as before, but the RHS changes.

Example Problem

First, swapping x and y gives us that

$$f(y)f(x) = y + f(x) + 57,$$

so the LHS remains the same as before, but the RHS changes. Now, equating the two RHS gives us that

$$x + f(y) + 57 = y + f(x) + 57.$$

By separation of variables, it follows that

$$f(x) - x = f(y) - y = c$$

for some $c \in \mathbb{R}$, and so $f(x) = x + c$.

Example Problem

Substituting $f(x) = x + c$ into our original equation gives us that

$$\begin{aligned}(x + c)(y + c) &= x + y + c + 57 \\ \implies xy + cx + cy + c^2 &= x + y + c + 57\end{aligned}$$

which is impossible by equating coefficients of xy on both sides. So no such f satisfying the equation exists.

Small tricks

Involutions

Definition

We say a function f is an involution if

$$(f \circ f)(x) = f(f(x)) = x, \quad \forall x,$$

that is, f is its own inverse function.

Involutions

Definition

We say a function f is an involution if

$$(f \circ f)(x) = f(f(x)) = x, \quad \forall x,$$

that is, f is its own inverse function.

How does this help us?

Involutions

If we spot an involution, which will typically be of the form $f(x) = \frac{ax+b}{cx+d}$ for constants a, b, c, d , then we can make the substitution $x \mapsto f(x)$ to help us solve the equation.

Involutions

If we spot an involution, which will typically be of the form $f(x) = \frac{ax+b}{cx+d}$ for constants a, b, c, d , then we can make the substitution $x \mapsto f(x)$ to help us solve the equation.

Example

Find all functions $f : \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ such that

$$x - f(x) = 5f\left(\frac{x}{x-1}\right).$$

Involutions

We can spot that $\frac{x}{x-1}$ is an involution, and so substituting $\frac{x}{x-1}$ for x gives us the equation

$$\frac{x}{x-1} - f\left(\frac{x}{x-1}\right) = 5f(x).$$

Involutions

We can spot that $\frac{x}{x-1}$ is an involution, and so substituting $\frac{x}{x-1}$ for x gives us the equation

$$\frac{x}{x-1} - f\left(\frac{x}{x-1}\right) = 5f(x).$$

As a result, we can rearrange for $f\left(\frac{x}{x-1}\right)$ in our two equations to get that

$$f\left(\frac{x}{x-1}\right) = \frac{1}{5}(x - f(x)) = \frac{x}{x-1} - 5f(x),$$

Involutions

We can spot that $\frac{x}{x-1}$ is an involution, and so substituting $\frac{x}{x-1}$ for x gives us the equation

$$\frac{x}{x-1} - f\left(\frac{x}{x-1}\right) = 5f(x).$$

As a result, we can rearrange for $f\left(\frac{x}{x-1}\right)$ in our two equations to get that

$$f\left(\frac{x}{x-1}\right) = \frac{1}{5}(x - f(x)) = \frac{x}{x-1} - 5f(x),$$

and so

$$f(x) = \frac{5}{24} \left(\frac{x}{x-1} - \frac{x}{5} \right).$$

Indeed this is a solution if we plug it back in and check!

Associativity

Proposition / Definition

Composition of functions is associative, meaning that

$$(f \circ g) \circ h = f \circ (g \circ h)$$

for functions f, g, h .

Associativity

Proposition / Definition

Composition of functions is associative, meaning that

$$(f \circ g) \circ h = f \circ (g \circ h)$$

for functions f, g, h .

How does this help us?

Associativity

Example

Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$f(f(x)) = x + 1$$

for all integers x .

Associativity

Let $g(x) = f(f(x)) = x + 1$, then applying f to both sides of the original equation we get that

$$f \circ f \circ f(x) = f(x + 1).$$

Associativity

Let $g(x) = f(f(x)) = x + 1$, then applying f to both sides of the original equation we get that

$$f \circ f \circ f(x) = f(x + 1).$$

Now using the associativity of function composition, we get that

$$\begin{aligned} f \circ (f \circ f)(x) &= f \circ g(x) \\ &= f(x + 1), \end{aligned}$$

$$\begin{aligned} (f \circ f) \circ f(x) &= g(f(x)) \\ &= f(x) + 1, \end{aligned}$$

and so we get that $f(x + 1) = f(x) + 1$.

Associativity

Since $f(x+1) - f(x) = 1$ for all integers x , it follows that f is linear, and so $f(x) = x + c$ for some integer c . As a result, we get that

$$f(f(x)) = x + 2c = x + 1$$

and so $c = \frac{1}{2}$, which is a contradiction. Hence no functions exist.

Appendix

Feedback

Please give us feedback and let us know what you would like to see in future workshops.

